No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

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Based on: Balcan, Blum, Haghtalab, Procaccia (EC 2015)

Outline

1. Literature review and motivation
2. Problem setup
3. Proof, algorithm, and results

Stackelberg Security Games (SSGs)

- A **defender** allocates limited resources to protect *n targets* against an **attacker**.
- Real deployments: US Coast Guard, LAX airport patrols, Federal Air Marshals, wildlife anti-poaching (Tambe 2012; Yang et al. 2014).
- Conceptually: **defender commits first**, attacker observes the committed (mixed) strategy and best-responds.

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The classical assumption

A single, perfectly known attacker utility function. The optimal commitment is then computable by an LP / MILP.

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- In reality, the defender faces **many possible attacker types** whose utility functions are not known in advance.

What the literature says about attacker uncertainty

- **Robust optimization** (Pita et al. 2010): plan against worst-case payoffs.
- **Bayesian SSG** (Kiekintveld et al. 2011; Marecki et al. 2012): assume a *prior* over attacker types.
- **Learning a single unknown type** (Letchford et al. 2009; Blum, Haghtalab, Procaccia 2014): query-efficient algorithms.
- **Online learning against an adversarial sequence of types** (Balcan, Blum, Haghtalab, Procaccia, EC 2015) – **our base paper**.

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Why Balcan et al. (2015) is the right starting point

- No prior over attacker types – adversarial sequence from a *known* set K.
- Sublinear regret bound, with matching information-theoretic lower bound.
- Open problem they leave: *what if K is not even known?*

Sequential vs. Simultaneous Security Games

Simultaneous

- Attacker does *not* see $\mathbf{p}_t$.
- Loss is **linear in p** $\Rightarrow$ standard online linear optimization.
- Off-the-shelf algorithms give $O(\sqrt{T \log n})$ regret.

Sequential (Stackelberg)

- Attacker sees $\mathbf{p}_t$ and best-responds $b_{a_t}(\mathbf{p}_t)$.
- Loss is **non-linear, non-convex, discontinuous** in $\mathbf{p}_t$.
- Needs extreme-point machinery (Balcan et al.).

Why we picked the sequential model

Sequential Stackelberg is **strictly harder for the defender**: the attacker can exploit any exposed corner of $\mathbf{p}_t$, while in the simultaneous model it has to guess. A bound proved for sequential is a **worst-case** bound that automatically covers the simultaneous case.

Where our project sits

- Balcan et al. prove: if K with $|K| = k$ is *known*, there is an online algorithm with regret $O(\sqrt{Tn^2 k \log(nk)})$ against any oblivious adversarial sequence from K.
- They also prove a **matching lower bound**: if $|K|$ is *unrestricted* (up to 2^{T+1}), every online algorithm incurs $\Omega(T)$ regret.

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What if K is unknown a priori, but $|K| \leq K_{\max}$?

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What if K is unknown a priori, but $|K| \leq K_{\max}$?

- A genuinely open middle ground between "known finite set" and "no bound".
- We focus on the *full-information* feedback setting to isolate the *information* gap from the partial-feedback one.

Game primitives (from Balcan et al., 2015)

Stackelberg Security Game

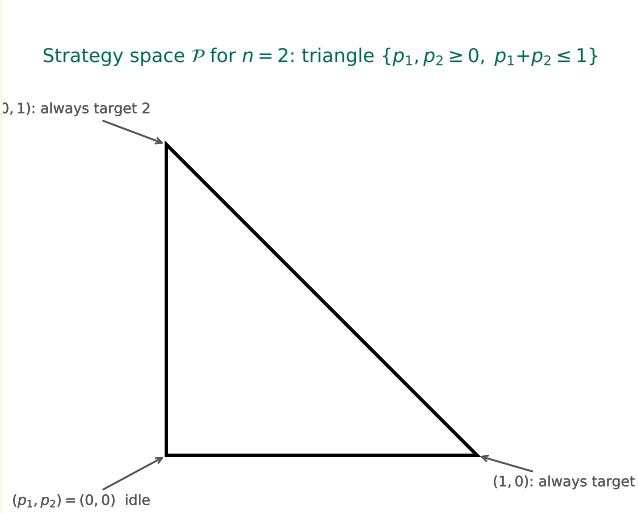
- Targets $N = \{1, \ldots, n\}$; defender has R resources, schedules $\mathcal{D} \subseteq 2^N$.
- Pure strategies: valid resource-to-schedule assignments. Mixed strategy $\mathbf{s} \in \Delta_m$.
- Induced **coverage probability vector** $\mathbf{p} = M\mathbf{s} \in [0,1]^n$; $p_i =$ prob. target i is defended.
- $\mathcal{P} \subseteq [0,1]^n$: the (convex) polytope of valid coverage vectors.

Defender utilities (variables defined before use)

- $u_d^c(i) \in [-1, 1]$: defender's payoff when target i is *attacked and covered*.
- $u_d^u(i) \in [-1, 1]$: defender's payoff when target i is *attacked and uncovered*.
- Expected defender utility when target i is attacked under coverage $\mathbf{p}$:

$$U_d(i, \mathbf{p}) \;=\; u_d^c(i)\, p_i \;+\; u_d^u(i)\,(1 - p_i).$$

Strategy space $\mathcal{P}$ for $n = 2$



- With $n = 2$ targets and one resource (which may be idle), the coverage space is the triangle

$$\mathcal{P} = \{(p_1, p_2) : p_1 \geq 0,\ p_2 \geq 0,\ p_1 + p_2 \leq 1\}.$$

- Vertex $(1, 0)$: always cover target 1.
- Vertex $(0, 1)$: always cover target 2.
- Vertex $(0, 0)$: resource is idle (no target defended).
- $\mathcal{P}$ is a convex polytope – the same 2D triangle used in Balcan et al., Figure 1.

Attacker types and best response

- An attacker **type** α is specified by utilities $u^c_\alpha(i), u^u_\alpha(i) \in [-1, 1]$:

$$U_\alpha(i, \mathbf{p}) \;=\; u^c_\alpha(i)\, p_i \;+\; u^u_\alpha(i)\,(1 - p_i).$$

- Observing $\mathbf{p}$, type α **best-responds**:

$$b_\alpha(\mathbf{p}) \;=\; \arg\max_{i \in N} U_\alpha(i, \mathbf{p}).$$

- Per-round payoff to the defender when type α attacks:

$$U_d\big(b_\alpha(\mathbf{p}), \mathbf{p}\big).$$

The technical difficulty

The best-response operator $b_\alpha(\cdot)$ is piecewise constant in $\mathbf{p}$. Hence the defender's loss is **not** linear or continuous in $\mathbf{p}$.

Repeated game and regret

- Time horizon T is known. An **oblivious** adversary fixes a sequence $\mathbf{a} = a_1, \ldots, a_T \in K^T$ before the game starts.
- At each round t:
  1. Defender commits $\mathbf{p}_t \in \mathcal{P}$.
  2. Attacker a_t plays $b_{a_t}(\mathbf{p}_t)$; defender receives $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$.
  3. **Full information feedback**: defender observes a_t (and, first time α is seen, its full utility).
- Benchmark – **best fixed coverage in hindsight**:

$$\mathbf{p}^* = \arg\max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T} U_d\big(b_{a_t}(\mathbf{p}), \mathbf{p}\big).$$

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Expected regret

$$\mathrm{Regret}(T) = \sum_{t=1}^{T} U_d\big(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*\big) - \mathbb{E}\Big[\sum_{t=1}^{T} U_d\big(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t\big)\Big].$$

Balcan, Blum, Haghtalab, Procaccia (2015) – known K

Theorem 5.1 (Balcan et al.)

If K with $|K| = k$ is *known* upfront and feedback is full information, there is an online algorithm achieving

$$\mathbb{E}[\mathrm{Regret}(T)] \;\leq\; O\left(\sqrt{T \cdot n^2 \cdot k \cdot \log(nk)}\right).$$

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- Proof technique: partition $\mathcal{P}$ into finitely many best-response regions, enumerate the **ϵ-approximate extreme point set** $\mathcal{E}(K;\epsilon)$, run an online learning subroutine over $\mathcal{E}$.
- Lower bound (Thm. 7.1): unrestricted $|K| \Rightarrow \Omega(T)$ regret.

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Our extension

$|K|$ unknown a priori, but we are given a constant $K_{\max}$ with $|K| \leq K_{\max}$. New types reveal their full utility when first observed.

Primer: half-spaces and convex polytopes

- A **half-space** in $\mathbb{R}^n$ is $\{\mathbf{p} : \mathbf{a}^\top \mathbf{p} \leq b\}$.
- A **convex polytope** is a bounded intersection of finitely many half-spaces.
- For each attacker type α and target i,

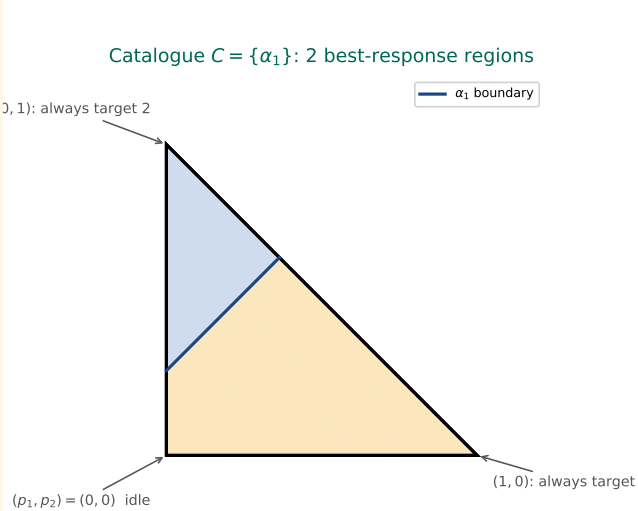
$$\mathcal{P}_i^\alpha \;=\; \{\mathbf{p} \in \mathcal{P} \;:\; b_\alpha(\mathbf{p}) = i\}$$

  is itself a convex polytope (defined by the pairwise linear inequalities $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$).
- For a response profile $\sigma : C \to N$,

$$\mathcal{P}_\sigma(C) \;=\; \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha$$

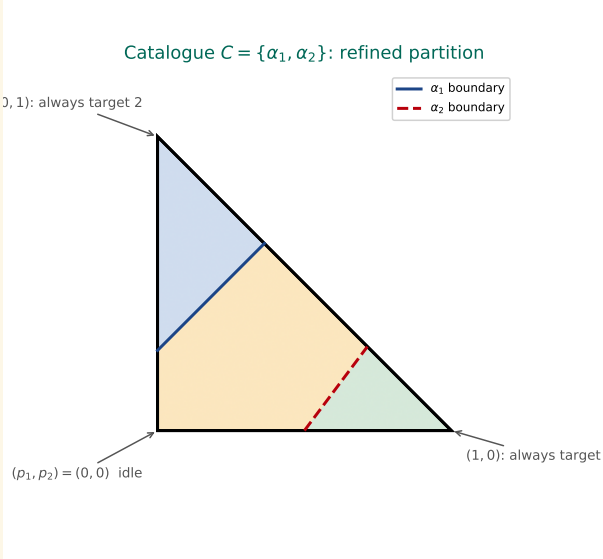
  is again a convex polytope.

A concrete example for $n = 2$: one best-response cut



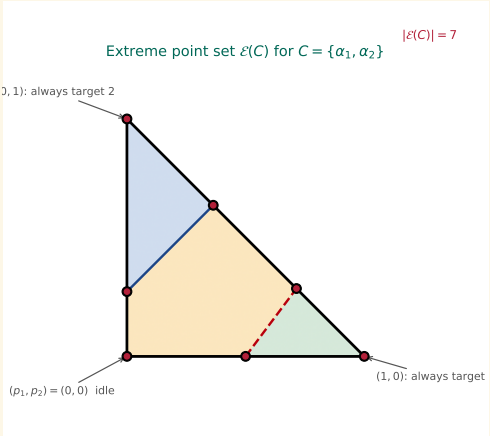
- With $n = 2$ and an idle option, $\mathcal{P}$ is the triangle $\{p_1, p_2 \geq 0,\ p_1 + p_2 \leq 1\}$.
- For one attacker α_1, $U_{\alpha_1}(1, \mathbf{p}) = U_{\alpha_1}(2, \mathbf{p})$ is a single line in (p_1, p_2)-space – one **best-response boundary**.
- On one side of the line α_1 attacks target 1; on the other, target 2.
- $\mathcal{P}$ is split into **2 convex polytopes**.

Adding more types refines the partition



- Each new type α adds $\binom{n}{2}$ linear boundaries ($= 1$ line when $n = 2$).
- Existing polytopes get **sliced**, never lost; the refined partition is still a collection of convex polytopes.
- Within any $\mathcal{P}_\sigma(C)$, the best response of every $\alpha \in C$ is fixed $\Rightarrow$ the defender's payoff is **linear** in $\mathbf{p}$ on that region.

The $\mathcal{E}(C)$ set of extreme points

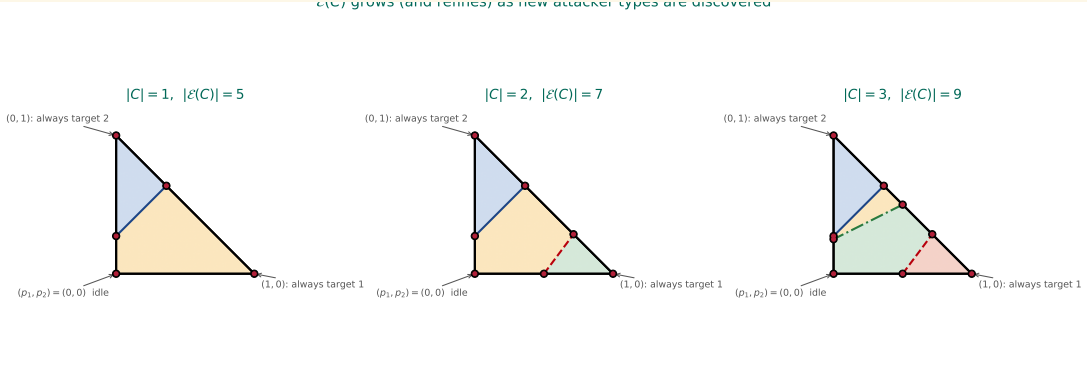


- Within each $\mathcal{P}_\sigma(C)$, the defender's payoff is linear in $\mathbf{p} \Rightarrow$ its maximum is attained at a **vertex** of $\mathcal{P}_\sigma(C)$.
- Define $\mathcal{E}(C;\epsilon) =$ the ϵ-approximate vertices of all $\mathcal{P}_\sigma(C)$.
- Key facts (Balcan et al., Lemmas 4.3, 4.4):

$$\max_{\mathbf{p}\in\mathcal{E}(C)} \sum_t U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p}\in\mathcal{P}} \sum_t U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon T$$

$$|\mathcal{E}(C;\epsilon)| \leq O\big((2^n + |C|n^2)^n \cdot n^{|C|}\big)$$

$\mathcal{E}(C)$ grows as new types are discovered



- Each new type adds a best-response line and hence new vertices.
- $|\mathcal{E}(C_i)|$ is bounded uniformly by $N_{\max} := |\mathcal{E}(C; \epsilon)|$ at $|C| = K_{\max}$.
- The analysis works on the *current* $\mathcal{E}(C)$ at each round – no need to anticipate future types.

Our main theorem

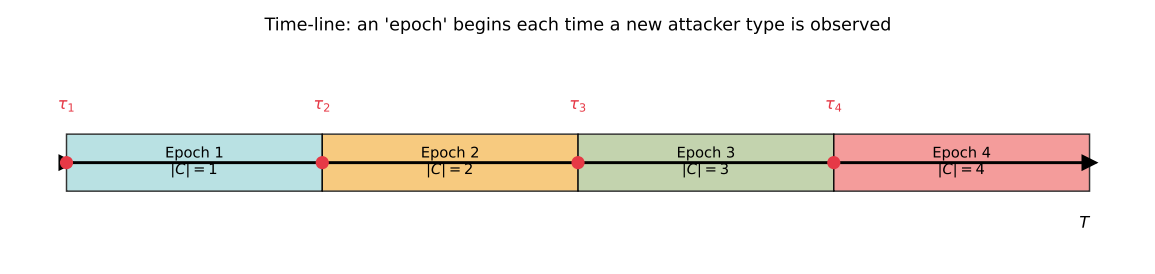
Theorem (unknown K, full information)

Let K be an unknown set of attacker types with $|K| \leq K_{\max}$. For full-information feedback and any oblivious adversarial sequence $\mathbf{a} \in K^T$, there is an online algorithm whose expected regret satisfies

$$\mathbb{E}[\mathrm{Regret}(T)] \;\leq\; O\Big(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\Big).$$

- Same form as Balcan et al. Theorem 5.1, with k replaced by $K_{\max}$.
- For $K_{\max} = O(1)$ the rate matches the known-K rate up to log factors.
- Instance-adaptive: if only $K' < K_{\max}$ types actually appear, the bound tightens to $O(\sqrt{TK'(n^2 + K_{\max} \log n)})$.

Proof idea in one picture



1. Split $[1, T]$ into **epochs**: epoch i starts right after the i-th new type is discovered and ends before the $(i+1)$-st.
2. Within an epoch, the catalogue C_i is fixed; run an online learning subroutine on $\mathcal{E}(C_i)$.
3. Apply Balcan et al.'s within-epoch analysis (regret bound + Lemma 4.3) to each epoch.
4. Sum over at most $K_{\max}$ epochs using **Cauchy–Schwarz**.
5. Absorb $K' \leq K_{\max}$ *discovery rounds* into a $2K_{\max}$ additive term.

Proof – within-epoch regret

Online learning black-box (Balcan et al. Proposition 1)

For any fixed expert set $\mathcal{E}_i$ and any L_i rounds, there is an online learning algorithm whose expected regret satisfies

$$\max_{\mathbf{p}\in\mathcal{E}_i} \sum_{t\in \mathrm{Ep.}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \; - \; \mathbb{E}\left[\sum_{t\in \mathrm{Ep.}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \; \leq \; c_1\sqrt{L_i \log|\mathcal{E}_i|}.$$

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Approximation by extreme points (Lemma 4.3)

Inside epoch i the sequence only uses types in C_i, so

$$\max_{\mathbf{p}\in\mathcal{E}_i} \sum_{t\in \mathrm{Ep}_{\cdot i}} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \;\ge\; \sum_{t\in \mathrm{Ep}_{\cdot i}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i.$$

Proof – combining the two within an epoch

Chaining the two inequalities from the previous slide gives, for every epoch i:

$$\mathbb{E}\left[\sum_{t \in \mathsf{Ep}_{\cdot i}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\geq\; \sum_{t \in \mathsf{Ep}_{\cdot i}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; 2\epsilon L_i \;-\; c_1\sqrt{L_i \log |\mathcal{E}_i|}.$$

- The left-hand side is the algorithm's expected utility on epoch i.
- The right-hand side compares against the global benchmark $\mathbf{p}^*$ restricted to epoch i.
- The $2\epsilon L_i$ term is the Lemma-4.3 approximation loss.
- The $c_1\sqrt{L_i \log |\mathcal{E}_i|}$ term is the within-epoch regret of the online learning subroutine.

Proof – summing epochs with Cauchy–Schwarz

Summing the per-epoch inequality over $i = 1, \ldots, K'$, using $\sum_i L_i \leq T$ and a discovery-round term $2K_{\max}$:

$$\mathbb{E}[\mathrm{Regret}(T)] \;\leq\; 2K_{\max} \;+\; 2\epsilon T \;+\; c_1\sqrt{\log N_{\max}}\sum_{i=1}^{K'}\sqrt{L_i}.$$

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By Cauchy–Schwarz,

$$\sum_{i=1}^{K'}\sqrt{L_i} \;\leq\; \sqrt{K' \cdot \textstyle\sum_i L_i} \;\leq\; \sqrt{K_{\max}\, T}.$$

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By Cauchy–Schwarz,

$$\sum_{i=1}^{K'}\sqrt{L_i} \;\leq\; \sqrt{K' \cdot \textstyle\sum_i L_i} \;\leq\; \sqrt{K_{\max}\, T}.$$

Plug $\epsilon = 1/\sqrt{T}$ and $\log N_{\max} = O(n^2 + K_{\max}\log n)$ to obtain

$$\mathbb{E}[\mathrm{Regret}(T)] \;\leq\; O\Big(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max}\log n)}\Big). \qquad \square$$

Regime in which the bound is meaningful

- The bound is sublinear in T iff $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$.
- Equivalently,

$$K_{\max} \;=\; o\left(\min\left\{\frac{T}{n^2},\, \sqrt{\frac{T}{\log n}}\right\}\right).$$

- $K_{\max} = O(1)$: $\tilde{O}(n\sqrt{T})$ for any $n \leq T^{1/2}$.
- $K_{\max} = \Theta(n^c)$: sublinear as long as $T = \omega(n^{c+2})$.
- For $K_{\max} \gtrsim T/n^2$ the upper bound becomes trivial – matches Balcan et al.'s linear-regret lower bound on the unbounded side.

The Grow-Catalogue algorithm

Algorithm (full information)

1. $C \leftarrow \emptyset$, $\mathcal{E} \leftarrow \emptyset$, $\epsilon \leftarrow 1/\sqrt{T}$.
2. For $t = 1, \ldots, T$:
   - If $\mathcal{E} = \emptyset$, play uniform $\mathbf{p}_{\text{def}}$; else sample $\mathbf{p}_t \sim \mathbf{q}_t$ where $\mathbf{q}_t$ is the current Hedge distribution over $\mathcal{E}$.
   - Observe a_t.
   - **New type** ($a_t \notin C$): add a_t (and its utility) to C, **recompute** $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$, restart Hedge on the new $\mathcal{E}$.
   - Else: compute $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$ for all $\mathbf{p} \in \mathcal{E}$, update Hedge with anytime learning rate $\eta_{t_{\text{loc}}} = \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$.

- We implement two variants: **total recomputation** and **incremental** (reuse surviving extreme points' weights). Same regret, different wall-clock.
- The anytime learning rate gives a $\sqrt{L_i}$-per-epoch guarantee without knowing L_i in advance.

Experiment design

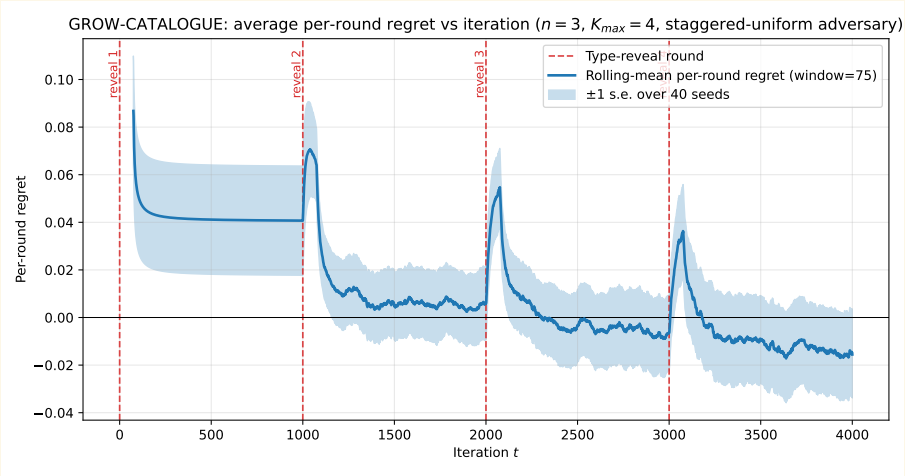
Staggered-uniform adversary

Partition $[0, T)$ into $K_{\max}$ phases of length $T/K_{\max}$. In phase i the attacker is drawn uniformly at random from $\{\alpha_1, \ldots, \alpha_i\}$:

- phase 1: only α_1;
- phase 2: $\{\alpha_1, \alpha_2\}$ each w.p. $\frac{1}{2}$;
- phase i: each of the i revealed types with prob. $\frac{1}{i}$.

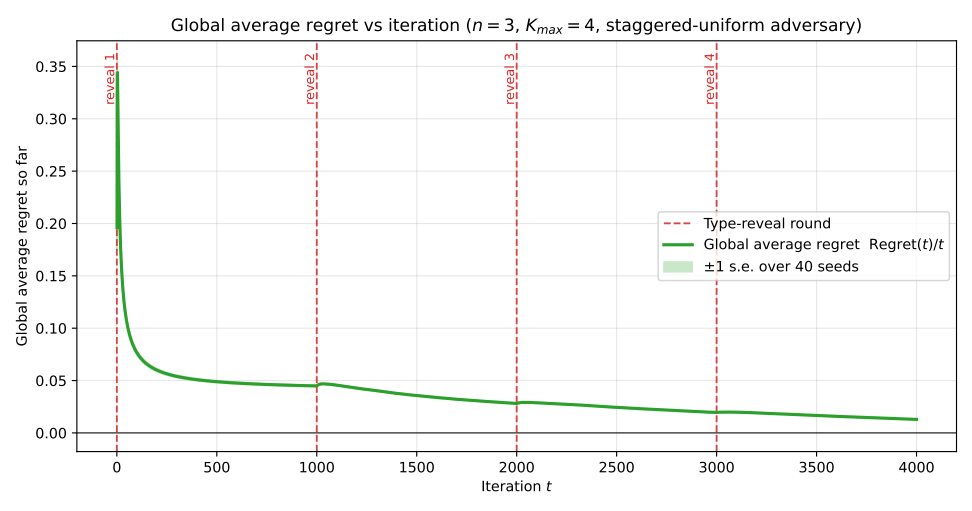
- $n = 3$, $K_{\max} = 4$, $T = 4000$, averaged over 40 random instances.
- Old types keep arriving after each reveal $\Rightarrow$ the benchmark $\mathbf{p}^*$ is a genuine compromise, not a specialist.
- Reveals happen at rounds 0, 1000, 2000, 3000.

Result 1: per-round regret vs iteration



- Clear **spikes** at each reveal round where a new type first appears, followed by re-convergence.
- Spikes shrink over time: more types already known $\Rightarrow$ each reveal is less disruptive.

Result 2: global average regret $\mathrm{Regret}(t)/t$



- Global average regret tends to 0 from above, consistent with the sublinear regret theorem.
- Tiny bumps at reveal rounds confirm the epoch structure.

Take-aways

- Sequential (Stackelberg) SSGs are the **harder** model – the right target for a worst-case bound.
- Unknown K with a known bound $K_{\max}$: we prove

$$\mathbb{E}[\mathrm{Regret}(T)] \;=\; O\Big(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\Big)\,,$$

  the same form as Balcan et al. with $k \to K_{\max}$.
- The bound is meaningful as long as $K_{\max} \cdot (n^2 + K_{\max} \log n) \ll T$.
- Experiments on a staggered-uniform adversary confirm sublinear regret with visible spikes at discovery rounds.

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Future directions

- **Partial-information** feedback (bandit version), needing adaptive barycentric spanners.
- **Adaptive adversary** under partial info (Balcan et al.'s explicit open problem).
- Tightening the logarithmic dependence on $K_{\max}$ via sleeping-experts techniques.

References I

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Thank you

Questions?